STABILIZABILITY OF FIRST-ORDER DYNAMICS IN SECOND-ORDER SYSTEMS

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Abstract. We study whether prescribed vector fields on compact smooth manifolds, viewed geometrically as sections of the tangent bundle, can be asymptotically stabilized by second-order control systems on the base manifold. Our class of second-order systems includes most Lagrangian systems, and we obtain both positive and negative results. The positive result asserts that, for fully actuated systems, the section corresponding to any smooth vector field can be made globally exponentially stable and normally hyperbolic. In particular, not only does each closed-loop trajectory converge to the prescribed section, but it also converges to a trajectory of the first-order dynamics generated by the corresponding vector field at an exponential rate. Thus, the closed-loop second-order system asymptotically reproduces the prescribed first-order dynamics. In contrast, the negative result asserts that, for underactuated systems on manifolds with nonzero Euler characteristic, sections corresponding to "almost all" smooth vector fields cannot even be locally asymptotically stabilized. This includes, in particular, all vector fields with only isolated zeros.

1. Introduction

We study general second-order control systems

$$\nabla_{\dot{q}}\dot{q} = F(u) \tag{1}$$

on a smooth (C^∞) Riemannian manifold (Q, κ), where ∇ is the Levi-Civita connection associated to the metric κ. See [1, 2, 3] for related standard details. In the general framework here, $F\colon U \to TQ$ is a map making the diagram

$$\begin{tikzcd} U \arrow[r, "F"] \arrow[d, "\pi_U"'] & TQ \arrow[d, "\pi_{TQ}"] \\ TQ \arrow[r, "\pi_{TQ}"'] & Q \end{tikzcd} \tag{2}$$

commute, where π_{TQ} is the tangent bundle projection and $\pi_U\colon U \to TQ$ is the control space projection. This framework extends a second-order version of Brockett's "bundle picture of control" [4, p. 15], [2, sec. 4.6], and includes all "regular" Lagrangian control systems [1, Def. 3.5.8] such as the standard class of "manipulator equations" in robotics with nondegenerate inertia tensor [5, Eq. 4.2.4], written locally as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau(q, \dot{q}, u). \tag{3}$$

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Remark 1. Since we work globally in this paper, we do not write redundant expressions such as "$F(q, \dot{q}, u)$" in (1) or elsewhere, as $u \in U$ contains all information about $\dot{q} = \pi_U(u)$, which in turn contains all information about $q = \pi_{TQ}(\dot{q})$.

This general framework is motivated by robotics [6], spacecraft [7], and other modern applications for which configuration spaces are non-Euclidean and/or the combined space of states and controls is not globally a product [8]. See [9] for related discussion in the context of similar frameworks for first-order systems, and [10, 11, 12] for examples of other works with related geometric-topological motivations.

For such general as well as classical second-order systems, a natural goal is to make the configuration q follow "reference dynamics" prescribed by a system of first-order ordinary differential equations

$$\dot{q} = X(q), \tag{4}$$

where X is a vector field on the configuration space Q. Here "reference dynamics" should be contrasted with the distinct notion of a "reference trajectory" $t \mapsto r(t) \in Q$ [13, 14, 15]. In seminal work, Koditschek showed that arbitrary first-order dynamics (4) can indeed be "embedded" within any "fully actuated" second-order system, at least in the classical Euclidean space context [16, sec. 3.3]. However, he also raised important concerns including that the closed-loop second-order system might not asymptotically reproduce dynamical properties of the prescribed first-order system.

The first main result of this paper, Theorem 1, is based on a coordinate-free version of Koditschek's method. It implies in particular that, if Q is compact, π_U is a smooth vector bundle, and F is a smooth fiberwise surjective linear map (see Proposition 1), then the image $X(Q) \subset TQ$ of an arbitrary smooth vector field X on Q can be made a globally exponentially stable and normally attracting invariant manifold (Definition 1) for the closed-loop dynamics on TQ resulting from substituting some smooth feedback law $u\colon TQ \to U$ into (1). Normal attractivity implies the crucial property that, not only does each closed-loop trajectory $t \mapsto \dot{q}(t)$ converge to $X(Q)$, but also $t \mapsto q(t)$ converges to a trajectory of (4) at an exponential rate [17, Thm 2], [18, Thm 4.1]. Thus, the closed-loop second-order system asymptotically reproduces the prescribed first-order reference dynamics (4), resolving the aforementioned concern in our setting (see Remark 4). See Figure 1 for an illustration.

While the first main result provides a positive conclusion for fully actuated systems, the second main result of this paper, Theorem 2, provides a negative conclusion for underactuated systems. It asserts in particular that, if Q is compact with nonzero Euler characteristic and the image $F(U) \subset TQ$ of F is contained in a regular distribution (vector subbundle) of positive corank, then the image $X(Q)$ of "almost every" smooth vector field X on Q cannot be made even locally asymptotically stable by feedback law $u\colon TQ \to U$ such that $F \circ u$ is smooth or even locally Lipschitz. Furthermore, the image $F(U)$ actually need only be contained in a regular distribution over a neighborhood of the zeros of X, broadening the applicability of Theorem 2.[1] To be more precise about the class of vector fields X to which the result applies, it includes all locally Lipschitz vector fields X having only finitely

[1]For example, there are no globally-defined continuous regular distributions of positive corank on even-dimensional spheres [19, problem 9-C], but Theorem 2 can still be applied to second-order control systems (1) on such spheres by considering locally-defined distributions, as in Example 2.

many zeros (see Remark 6). (The set of all C^k such vector fields contains an open and dense subset of the space of all C^k vector fields on Q with the C^k topology for any $k \in \mathbb{N}_{\geq 1} \cup \{\infty\}$ [20, Thm 2.3.2].)

The proof of the second main result is based on a generalization and strengthening of Brockett's necessary condition for stabilizability of points in Euclidean spaces [21, Thm 1.(iii)] to a necessary condition for stabilizability of general compact sets in manifolds introduced by the author and Koditschek [22, Thm 3.2] (see also [9, Thm 1]). See [23] for a broad survey of such topological obstructions to stability and stabilization, and [9] for a narrower review.

The remainder of this paper is organized as follows. General preliminary definitions are in section 2. The ("positive") first main result is stated and proved in section 3. The ("negative") second main result is stated in section 4, and it is proved in section 5. A concluding discussion is in section 6.

2. Preliminaries

A **section** of a map $\pi\colon A \to B$ is a right inverse of π, that is, a map $\sigma\colon B \to A$ satisfying $\pi \circ \sigma = \mathrm{id}_B$. If A and B are topological spaces, we let $\Gamma^0(\pi)$ denote the set of C^0 (continuous) sections of π. If A and B are smooth manifolds, we let $\Gamma^k(\pi)$ denote the set of C^k sections of π for $k \in \mathbb{N}_{\geq 0} \cup \{\infty\}$, where a map is C^k if it has C^0 partial derivatives of all orders less than $k+1$ in local coordinates.

Given any smooth manifold M, subset $V \subset M$, and point $v \in V$, let $\pi_{TM}\colon TM \to M$ be the tangent bundle projection of M, $T_V M = \pi_{TM}^{-1}(V)$, and $T_v M = T_{\{v\}} M$. We let $\mathfrak{X}^k(M) = \Gamma^k(\pi_{TM})$ denote the set of C^k vector fields on M for $k \in \mathbb{N}_{\geq 0} \cup \{\infty\}$. Given a C^1 map $g\colon M \to W$ into another manifold W, let $Tg\colon TM \to TW$ be the tangent map [24, p. 18], $T_V g = Tg|_{T_V M}\colon T_V M \to T_{g(V)} W$, and $T_v g = T_{\{v\}} g = Tg|_{T_v M}\colon T_v M \to T_{g(v)} W$.

Throughout this paper, the **configuration space** Q is a compact smooth manifold with smooth Riemannian metric κ, and ∇ is the associated Levi-Civita connection. (However, less differentiability would suffice.) The **control space** U is a topological space and the **control space projection** $\pi_U\colon U \to TQ$ is a C^0 map. Given any subset $V \subset TQ$ and point $v \in TQ$, we let $U_V = \pi_U^{-1}(V)$ and $U_v = U_{\{v\}}$.

In addition to vector fields on Q, we will need to consider vector fields on TQ such as the **geodesic spray** $S \in \mathfrak{X}^\infty(TQ)$ of the Riemannian metric κ [25, sec. 22.6]. (Geodesics $\mathbb{R} \to Q$ for κ are π_{TQ}-projections of trajectories $\mathbb{R} \to TQ$ of S.) Moreover, in addition to TQ, we will need to consider the tangent bundle $T^2Q = T(TQ)$ of TQ with tangent bundle projection $\pi_{T^2Q}\colon T^2Q \to TQ$. We denote by $V(TQ)$ the **vertical bundle** $\ker T\pi_{TQ} \subset T^2Q$,

$$TQ \times_Q TQ = \{(v,w) \in TQ \times TQ \colon \pi_{TQ}(v) = \pi_{TQ}(w)\}$$

the fiber product of π_{TQ} with itself (equivalently, the pullback of π_{TQ} along itself, which coincides with the Whitney sum $TQ \oplus TQ$ as a manifold but not as a vector bundle), and $\mathrm{vl}\colon TQ \times_Q TQ \to V(TQ)$ the **vertical lift** given by

$$\mathrm{vl}(v,w) = \frac{d}{dt}(v + tw)|_{t=0} \tag{5}$$

for all $(v,w) \in TQ \times_Q TQ$ [25, p. 107].

3. Stabilizability in the fully actuated case

3.1. **Normally attracting invariant manifolds.** In this subsection, we give precise definitions of k-normal attractivity and k-center bunching. The former corresponds to [26, sec. 2.1], [18, Prop. 2.3], which is a natural version for flows of the classical definition of eventual relative k-normal hyperbolicity for diffeomorphisms in the special case of a trivial unstable bundle [18, sec. 1] (see also [27, 28]). The latter corresponds to a hypothesis in [26, Cor. 2], which in turn corresponds to a hypothesis in [29, Thm 5] reformulated in notation closer to [30, Thm B] in the case $k = 1$. We first recall standard necessary ingredients.

Let V be a C^1 embedded submanifold of M [24, pp. 13, 21]. We let $N_V M = T_V M/TV$ denote the algebraic normal bundle of V in M [24, p. 96] and let $N_v M = T_v M/T_v V \subset N_V M$ for $v \in V$. If $g\colon M \to M$ is a C^1 map satisfying $g(V) \subset V$, then $Tg(TV) \subset TV$, so there is a canonical fiberwise linear map $N_V g\colon N_V M \to N_V M$.

Given a fiberwise linear map $\|L\|\colon E_1 \to E_2$ between topological vector bundles $\pi^i\colon E^i \to B^i$ with C^0 fiberwise norms $\|\cdot\|$ [24, sec. 4.2], let $E^i_b = (\pi^i)^{-1}(b)$ and

$$\|L|_{E^1_b}\| = \max_{\|e\|=1} \|L|_{E^1_b}(e)\| \quad \text{and} \quad \llbracket L|_{E^1_b}\rrbracket = \min_{\|e\|=1} \|L|_{E^1_b}(e)\|$$

denote the induced norm and minimum norm of $L|_{E^1_b}$.

Definition 1. Let V be a compact C^1 invariant manifold for a C^1 flow Φ on a smooth manifold M, and let Ψ be the flow on $N_V M$ given by $\Psi^t = N_V \Phi^t$. Fix any C^0 fiberwise norms $\|\cdot\|$ on TV and $N_V M$. Given $k \in \mathbb{N}_{\geq 1}$, we say that V is a **k-normally attracting invariant manifold (k-NAIM)** if there are constants $a, b > 0$ such that, for all $v \in V$ and integers $0 \leq i \leq k$,

$$\|\Psi^t|_{N_v M}\| \leq b e^{-at} \llbracket T\Phi^t|_{T_v V}\rrbracket^i \quad \text{for all } t \geq 0.$$

We say that V is a **normally attracting invariant manifold (NAIM)** if V is a 1-NAIM. We say that a NAIM $V \subset M$ is **k-center bunching** if there are constants $a, b > 0$ such that, for all $v \in V$ and integers $0 \leq i \leq k$,

$$\|T_v\Phi^t|_{T_v V}\|^i \|\Psi^t|_{N_v V}\| \leq b e^{-at} \llbracket T_v\Phi^t|_{T_v V}\rrbracket \quad \text{for all } t \geq 0.$$

Remark 2. Any pair of C^0 fiberwise norms on a vector bundle over a compact base are uniformly equivalent, so Definition 1 does not depend on the choice of norms.

Remark 3. If $k \in \mathbb{N}_{\geq 1}$, V is a k-NAIM for a C^k flow Φ, and Θ is another flow close to Φ in the compact-open (weak) C^k topology [24, sec. 2.1], then a classical fundamental result asserts that Θ has a unique k-NAIM V' that is C^k-close to V [18, Thm 4.1(f)]. Moreover, it is straightforward to show that V is k-center bunching for Φ if and only if there is $\tau > 0$ such that, for all $v \in V$ and integers $0 \leq i \leq k$,

$$\|T_v\Phi^\tau|_{T_v V}\|^i \|\Psi^\tau|_{N_v V}\| < \frac{1}{2} \llbracket T_v\Phi^\tau|_{T_v V}\rrbracket, \tag{6}$$

and similarly for V' and Θ. This is clearly a C^1-open condition on flows, so the k-NAIM V' for Θ will also be k-center bunching if Θ is sufficiently C^k-close to Φ.

3.2. **Results.** Theorem 1 incorporates the following assumption.

Assumption 1. For any C^k map $G\colon TQ \to TQ$ satisfying $G \circ \pi_{TQ} = \pi_{TQ}$, there is $u \in \Gamma^k(\pi_U)$ such that $F \circ u = G$.

In particular, Assumption 1 holds if π_U is isomorphic to the pullback of the bundle π_{TQ} along π_{TQ} [24, p. 97]. However, a more general sufficient condition is the following.

Proposition 1. If π_U is a C^k vector bundle and F is a C^k fiberwise surjective linear map, then Assumption 1 holds.

Proof. Let $\pi_P\colon P \to TQ$ be the C^k vector bundle given by the pullback of $\pi_{TQ}\colon TQ \to Q$ along π_{TQ}, and $p\colon P \to TQ$ be the associated C^k fiberwise linear isomorphism [24, p. 97]. By the associated universal property, there is a unique C^k fiberwise surjective linear map $\tilde{F}$ making

$$\begin{tikzcd} U \arrow[dr, dashed, "\tilde{F}"] \arrow[drr, bend left, "F"] \arrow[ddr, bend right, "\pi_U"'] & & \\ & P \arrow[r, "p"] \arrow[d, "\pi_P"'] & TQ \arrow[d, "\pi_{TQ}"] \\ & TQ \arrow[r, "\pi_{TQ}"'] & Q \end{tikzcd} \tag{7}$$

a commutative diagram. Since $p^{-1}(0_{TQ}) = 0_P$, the kernel

$$K := F^{-1}(0_{TQ}) = \tilde{F}^{-1}(p^{-1}(0_{TQ})) = \tilde{F}^{-1}(0_P) \subset U$$

is a C^k vector subbundle [24, p. 93], so there is a C^k vector subbundle $K^\perp \subset U$ satisfying $U = K \oplus K^\perp$ [24, p. 95]. Thus, $A := F|_{K^\perp}\colon K^\perp \to TQ$ is a C^k fiberwise linear isomorphism. Letting $A_v\colon K_v^\perp \to T_{\pi_{TQ}(v)}Q$ denote the restriction of A to $K_v^\perp := K^\perp \cap U_v$, we define $u \in \Gamma^k(\pi_U|_{K^\perp}) \subset \Gamma^k(\pi_U)$ by $u(v) = A_v^{-1}(G(v))$ and see that $F \circ u(v) = A_v(A_v^{-1}(G(v))) = G(v)$ as desired. □

Proposition 1 and Remark 1 should be kept in mind for the following statement of the first main result, a stabilization result with additional performance guarantees via an explicit feedback law for fully actuated systems.[2]

Theorem 1. Fix $k \in \mathbb{N}_{\geq 1}$ and let F satisfy Assumption 1. Then for any $X \in \mathfrak{X}^k(Q)$, C^k Riemannian metric ν on Q, and $\varepsilon > 0$, the closed-loop C^k dynamics determined by

$$\nabla_{\dot{q}}\dot{q} = F \circ u_\varepsilon(\dot{q}) \tag{8}$$

for any feedback law $u_\varepsilon \in \Gamma^k(\pi_U)$ satisfying

$$F \circ u_\varepsilon(v) = \nabla_v X - \frac{1}{\varepsilon}\kappa^\sharp \circ \nu^\flat(v - X \circ \pi_{TQ}(v)) \tag{9}$$

are complete and have $X(Q) \subset TQ$ as a globally exponentially stable invariant manifold. Moreover, $X(Q)$ is a k-NAIM and k-center bunching for all sufficiently small $\varepsilon > 0$.

[2]Here $\kappa^\flat, \nu^\flat\colon TQ \to T^*Q$ and $\kappa^\sharp, \nu^\sharp\colon T^*Q \to TQ$ are the standard **musical isomorphisms** with respect to the metrics κ and ν [1, p. 128].

Remark 4. Remarkably, the feedback law (9) is simply a coordinate-free version of Koditschek's classical formula in the Euclidean space setting [16, eq. (20)]. Following [16, Prop. 8], Koditschek articulates some important concerns:

> "Unfortunately, this result is much too weak to be useful. To begin with there is the assumption of completeness—i.e. that no finite escape trajectories result from the application of the control law. More fundamentally, the mere guarantee that $\mathcal{R}$ is a globally attracting set does not afford the conclusion that the behavior of trajectories starting away from $\mathcal{R}$ end up behaving at all like those which do originate in $\mathcal{R}$. In particular, if...[the dynamics]...is known to induce a positive limit set $\Omega \subset \mathcal{R}$, it is by no means clear that Ω is the positive limit set of the flow on the entire space."

(Here $\mathcal{R}$ corresponds to our $X(Q)$.) Theorem 1 completely addresses these concerns in our setting. To explain, let Φ_ε be the C^k flow on TQ determined by (8), (9) and Φ_X be the C^k flow of X on Q. First, Theorem 1 asserts that Φ_ε is (both forward and backward) complete for any $\varepsilon > 0$, resolving the completeness concern. Next, let $\varepsilon > 0$ be small enough as in the final sentence of Theorem 1. Then it follows that there is a C^{k-1} diffeomorphism $H_\varepsilon \colon TQ \to TQ$ conjugating Φ_ε to a C^{k-1} flow Θ_ε on TQ covering Φ_X such that 0_{TQ} is globally exponentially stable for Θ_ε [26, Cor. 2]. In particular,

$$\begin{tikzcd} TQ \arrow[r, "H"] & TQ \arrow[r, "\pi_{TQ}"] & Q \\ TQ \arrow[u, "\Phi_\varepsilon^t"] \arrow[r, "H"'] & TQ \arrow[u, "\Theta_\varepsilon^t"] \arrow[r, "\pi_{TQ}"'] & Q \arrow[u, "\Phi_X^t"'] \end{tikzcd}$$

is a commutative diagram for all $t \in \mathbb{R}$. It follows readily that, with respect to the natural diffeomorphic identification $X \colon Q \to X(Q)$, each Φ_ε-trajectory asymptotically coalesces with some Φ_X-trajectory at an exponential rate. This implies that all positive limit sets and, e.g., the chain recurrent sets of Φ_X and Φ_ε coincide under the natural identification. This is illustrated concretely in Example 1 and Figure 1. Of course, we have the technical advantage that Q is compact, unlike in Koditschek's classical Euclidean space setting.

Proof. First, since Q is compact, $v \mapsto \nu(v,v)$ attains a minimum $c > 0$ and maximum $C > 0$ on the κ-unit sphere bundle in TQ, so for all $v \in TQ$ we have

$$c\kappa(v,v) \leq \nu(v,v) \leq C\kappa(v,v). \tag{10}$$

Let $t \mapsto q(t) \in Q$ be any solution of (8), (9). Define $y(t) = \dot{q}(t) - X(q(t))$. Then (9) implies that

$$\begin{aligned} \nabla_{\dot{q}} y &= F \circ u_\varepsilon(\dot{q}) - \nabla_{\dot{q}} X = -\frac{1}{\varepsilon}\kappa^\sharp \circ \nu^\flat(\dot{q} - X(q)) \\ &= -\frac{1}{\varepsilon}\kappa^\sharp \circ \nu^\flat(y), \end{aligned} \tag{11}$$

so differentiating $t \mapsto \kappa(y(t), y(t))$ yields

$$\frac{d}{dt}\kappa(y,y) = 2\kappa(\nabla_{\dot{q}} y, y) = -\frac{2}{\varepsilon}\kappa(\kappa^\sharp \circ \nu^\flat(y), y) = -\frac{2}{\varepsilon}\nu(y,y).$$

This and (10) yield

$$-\frac{2C}{\varepsilon}\kappa(y,y) \leq \frac{d}{dt}\kappa(y,y) \leq -\frac{2c}{\varepsilon}\kappa(y,y),$$

so for all t in the domain of $t \mapsto \dot{q}(t)$ we have

$$e^{-(2C/\varepsilon)t}\kappa(y_0, y_0) \leq \kappa(y(t), y(t)) \leq e^{-(2c/\varepsilon)t}\kappa(y_0, y_0),$$

where $y_0 = y(0)$. Since $v \mapsto \kappa(v, v)$ has compact sublevel sets, it follows that the dynamics of (8), (9) are complete and $X(Q)$ is a globally exponentially stable invariant manifold.

The dynamics of (8), (9) is the flow of the vector field $Y_\varepsilon \in X^k(TQ)$ given by

$$Y_\varepsilon(v) = S(v) + \mathrm{vl}(v, \nabla_v X) - \frac{1}{\varepsilon}\mathrm{vl}(v, \kappa^\sharp \circ \nu^\flat(v - X \circ \pi_{TQ}(v))),$$

where S is the geodesic spray of κ and vl is the vertical lift (section 2).

Define a "regularized" vector field $\tilde{Y}_\varepsilon = \varepsilon Y_\varepsilon \in X^k(TQ)$. This has the nice property that, while $(\varepsilon, v) \mapsto Y_\varepsilon(v)$ is well-defined and C^k only on $(\mathbb{R} \setminus \{0\}) \times TQ$, $(\varepsilon, v) \mapsto \tilde{Y}_\varepsilon(v)$ is well-defined and C^k on all of $\mathbb{R} \times TQ$. Moreover, trajectories $t \mapsto v(t)$ of $\tilde{Y}_\varepsilon$ satisfy $v(t) = \dot{q}(\varepsilon t)$, where $t \mapsto \dot{q}(t)$ is the trajectory of Y_ε satisfying $v(0) = \dot{q}(0)$. Thus, for any $\varepsilon > 0$, $X(Q)$ is a k-NAIM and k-center bunching for Y_ε if and only if it is the same for $\tilde{Y}_\varepsilon$.

Note that $X(Q)$ is the set of zeros of the vector field

$$\tilde{Y}_0(v) = -\mathrm{vl}(v, \kappa^\sharp \circ \nu^\flat(v - X \circ \pi_{TQ}(v))).$$

Since κ, ν are given locally by symmetric positive definite matrices, it readily follows that the linearization of $\tilde{Y}_0$ at any such zero has $\dim Q$ zero eigenvalues and $\dim Q$ negative real eigenvalues. Thus, $X(Q)$ is a k-NAIM and k-center bunching for $\tilde{Y}_0$. Since $\tilde{Y}_\varepsilon \to \tilde{Y}_0$ in the compact-open C^k topology as $\varepsilon \to 0$, it follows that, for all sufficiently small $\varepsilon > 0$, $X(Q)$ is a k-NAIM and k-center bunching (see Remark 3) for the flow of $\tilde{Y}_\varepsilon$ and hence also the flow of Y_ε, i.e., the dynamics determined by (8), (9). □

Remark 5. We close this section by noting a connection with linearizing, or "Koopman", embeddings of nonlinear systems (see e.g. [31, 32, 33, 34, 35, 36, 37]). In the notation of Remark 4, $X \circ \pi_{TQ} \circ H_\varepsilon$ is a C^{k-1} **asymptotic phase map** in the sense of [38, sec. 2.3–2.4]. By [38, Thm 4], the existence of a C^{k-1} asymptotic phase map is necessary for the existence of a C^{k-1} embedding $h\colon TQ \to \mathbb{R}^n$ that linearizes the dynamics of (8), (9), i.e., $y = h(\dot{q})$ satisfies

$$\dot{y} = Ay$$

for some $n \in \mathbb{N}$ and $A \in \mathbb{R}^{n \times n}$. Moreover, the existence of a C^0 asymptotic phase map is necessary even for the existence of an injective C^0 map that linearizes the dynamics, and any such linearizing map is automatically a proper C^0 embedding [38, Thm 3].

Example 1. Let Q be the unit sphere $S^2 \subset \mathbb{R}^3$ with Riemannian metric $\kappa = \nu$ given by the round metric, namely, the restriction of the Euclidean inner product $\langle \cdot, \cdot \rangle$ to TS^2. Viewing TQ as an embedded submanifold of $\mathbb{R}^3 \times \mathbb{R}^3$ in the standard way [39, p. 50], let $X \in \mathfrak{X}^\infty(S^2)$ be given by

$$X(q_1, q_2, q_3) = (-q_2, q_1, 0). \tag{12}$$

Let $\Pi\colon T_{S^2}\mathbb{R}^3 \to TS^2$ be the fiberwise orthogonal projection

$$\Pi(w) = w - \langle w, q \rangle q$$

for $q \in S^2$ and $w \in T_q\mathbb{R}^3$. Then the Levi-Civita connection ∇ for κ satisfies

$$\begin{aligned}\nabla_v X &= \Pi(TX(v)) \\ &= D_q X \cdot v - \langle D_q X \cdot v, q\rangle q\end{aligned}$$

for $q \in S^2$ and $v \in T_q S^2$, where $D_q X \in \mathbb{R}^{3\times 3}$ is the Jacobian matrix of the map $\mathbb{R}^3 \to \mathbb{R}^3$ defined by the right side of (12), "$\cdot$" denotes matrix multiplication, and q, v are viewed as column vectors. Similarly,

$$\begin{aligned}\nabla_{\dot q}\dot q &= \Pi(\ddot q) = \ddot q - \langle \ddot q, q\rangle q \\ &= \ddot q + \langle \dot q, \dot q\rangle q\end{aligned}$$

for any C^2 curve $t \mapsto q(t) \in S^2$, where the third equality follows from twice differentiating the constant curve $t \mapsto \langle q(t), q(t)\rangle \equiv 1$ using the product rule. Thus, the closed-loop dynamics (8), (9) take the concrete form

$$\ddot q = D_q X \cdot v - \langle D_q X \cdot v, q\rangle q - \langle \dot q, \dot q\rangle q - \frac{1}{\varepsilon}(\dot q - X(q)). \tag{13}$$

Note that, since the first-order dynamics have no hyperbolicity, $X(Q)$ is a globally exponentially stable ∞-NAIM and ∞-center bunching (k-NAIM and k-center bunching for every finite k) for the second-order system (13) for *any* $\varepsilon > 0$. Taking $\varepsilon = 1.2$, Figure 1 shows solutions $t \mapsto q(t)$ for several initial conditions $(q(0), \dot q(0))$ converging to corresponding trajectories of the first-order dynamics generated by the vector field (12) as guaranteed (see Remark 4).

4. Non-stabilizability in the underactuated case

Theorem 2 incorporates the following assumption.

Assumption 2. There is a neighborhood $W \subset Q$ of $X^{-1}(0_{TQ})$, a C^0 regular distribution $\mathcal{D}$ on W, and $Y \in \mathfrak{X}^0(W)$ such that $F(U_{T_W Q}) \subset \mathcal{D}$ and $Y(W) \cap \mathcal{D} = \varnothing$.

Remark 6. If $X^{-1}(0_{TQ})$ is finite and there is a C^0 regular distribution $\mathcal{D}'$ of positive corank on a neighborhood $W' \subset Q$ of $X^{-1}(0_{TQ})$ such that $F(U_{T'_W Q}) \subset \mathcal{D}'$, then Assumption 2 holds for some $W \subset W'$ and $\mathcal{D} = \mathcal{D}'|_W$.

Remarks 1, 6 should be kept in mind for the following statement of the second main result, a non-stabilizability result for underactuated systems. For the statement, we say that (14) is **uniquely integrable** if there is a unique maximal trajectory $t \mapsto q(t)$ satisfying $\dot q(0) = v_0$ for any $v_0 \in TQ$.

Remark 7. By the Picard-Lindelöf theorem, (14) is automatically uniquely integrable if $F \circ u$ is locally Lipschitz (e.g., U is a smooth manifold and u, F are locally Lipschitz).

Theorem 2. Fix any $X \in \mathfrak{X}^0(Q)$. Assume that the Euler characteristic of Q is nonzero and Assumption 2 holds. Then for any $u \in \Gamma^0(\pi_U)$ such that

$$\nabla_{\dot q}\dot q = F \circ u(\dot q) \tag{14}$$

is uniquely integrable, $X(Q)$ is not an asymptotically stable invariant set for the dynamics determined by (14).

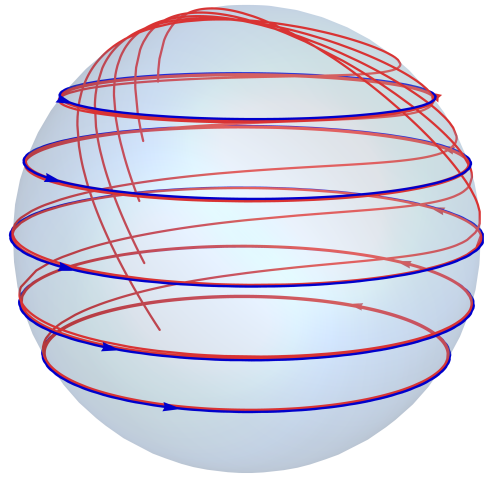

FIGURE 1. Illustration of Example 1. Shown in blue are several trajectories of the first-order reference dynamics generated by the vector field (12) on $Q = S^2$. Shown in red are several solutions to the closed-loop second-order system (13) for $\varepsilon = 1.2$. As explained in Example 1, here $X(S^2) \subset TS^2$ is an ∞-NAIM and ∞-center bunching for *any* value of $\varepsilon > 0$. Indeed, each red solution exponentially converges to a blue trajectory, and the closed-loop second-order system asymptotically reproduces the prescribed first-order reference dynamics, as guaranteed by NAIM theory (see Remark 4).

Example 2. Let $Q = S^2 \subset \mathbb{R}^3$, $\kappa = \nu$, and X be as in Example 1. However, viewing F as an $\mathbb{R}^3$-valued map, assume now that the first component of $F(u)$ is constrained to vanish whenever $q \in S^2$ is near the north and south poles constituting the zero set of X, so F takes values there in the smooth regular distribution $\mathcal{D}$ defined by

$$\mathcal{D}_q = \{v \in T_q S^2 \colon v_1 = 0\}.$$

Then Assumption 2 holds by Remark 6, so Theorem 2 implies that $X(S^2) \subset TS^2$ cannot even be made locally asymptotically stable for (14) for any locally Lipschitz feedback law $u \in \Gamma^0(\pi_U)$, let alone globally exponentially stable (and more) as in Example 1.

5. PROOF OF THEOREM 2

5.1. **The adversary theorem.** To prove Theorem 2, we will use an extension of Brockett's necessary condition for stabilizability of points in Euclidean spaces [21, Thm 1.(iii)] to one for stabilizability of general compact sets in manifolds introduced by the author and Koditschek [22, Thm 3.2] (see also [9, Thm 1]). In this subsection we recall the statement of this result for the special case of stabilizability of submanifolds.

The result uses a general framework for first-order control systems. It consists of a **state space** given by a smooth manifold M, a **control space** given by a

topological space U and a C^0 map $\pi_U \colon U \to M$, and a **control system** given by a C^0 map $f \colon U \to TM$ making the diagram

$$\begin{tikzcd} U \arrow[rr, "f"] \arrow[dr, "\pi_U"'] & & TM \arrow[dl, "\pi_{TM}"] \\ & M & \end{tikzcd} \tag{15}$$

commute. (We continue to use our global notations U and π_U in this subsection since no confusion will arise, as the only specific first-order control system considered later in this paper will have U and π_U coinciding with those in (1).)

A compact set $V \subset M$ is **stabilizable** if there is a **feedback law** $u \in \Gamma^0(\pi_U)$ for which the **closed-loop vector field** $f \circ u$ is uniquely integrable and has V as an asymptotically stable invariant set. Here $f \circ u$ is **uniquely integrable** if there is a unique maximal trajectory $t \mapsto x(t)$ satisfying $x(0) = x_0$ for any $x_0 \in M$.

Remark 8. Similarly to Remark 7, $f \circ u$ is automatically uniquely integrable if $f \circ u$ is locally Lipschitz (e.g., if U is a smooth manifold and u, F are locally Lipschitz).

Remark 9. This framework reduces to the (fiber) "bundle picture of control" [4, p. 15], [2, sec. 4.6] in the special case that $\pi_U \colon U \to M$ is a fiber bundle and $f \colon U \to TM$ is sufficiently smooth [4, p. 15], [2, §4.6]. If π_U is a trivial (product) bundle $\pi_U \colon U = M \times Y \to M$, then elements of U are of the form (x, u), so the expression "$f(x, u)$" makes sense. But in the general framework here, it is more appropriate to instead write expressions such as "$f(u)$" for $u \in U$ and "$f \circ u$" for $u \in \Gamma^0(\pi_U)$, keeping in mind that $u \in U$ contains all information about $x = \pi_U(u)$ (cf. Rem. 1).

The result below refers to the following reformulation [9, sec. 1.4] more closely resembling [21, Thm 1.(iii)] of a condition from [22, Thm 3.2]. The terminology is explained by interpreting it as the ability to "defeat" "adversaries" Y via the image of f somewhere intersecting that of Y.

Adversary condition. For any neighborhood $W_V \subset M$ of $V \subset M$ there is a neighborhood $W_0 \subset TM$ of the zero section such that, for any W_0-valued vector field Y over W_V, $Y(x) = f(u_x)$ for some $x \in W_V$, $u_x \in U_x$.

We elaborate on the Euler characteristic after stating the following special case of the result [22, Thm 3.2], [9, Thm 1].

Theorem 3. Let $V \subset M$ be homeomorphic to a compact smooth manifold with nonzero Euler characteristic. If V is stabilizable for (15), then the adversary condition holds.

Since V is homeomorphic to a compact smooth manifold, V admits a finite triangulation [40, Thm II.10.6] and the Euler characteristic of V is the usual alternating sum

$$\#(\text{vertices}) - \#(\text{edges}) + \#(\text{faces}) - \dots,$$

an integer. However, for the case of a general compact set $V \subset M$, the Euler characteristic instead needs to be defined using Čech(-Alexander-Spanier) cohomology for the more general version of Theorem 3 to hold. See [9, sec. 11.2] for a brief overview and [22, sec. 2] for further details.

5.2. **Proof.** In this section we prove two lemmas and then Theorem 2. We will use a specific case of the general first-order control system (15) given by the commutative diagram

$$\begin{tikzcd} U \arrow[rr, "f"] \arrow[dr, "\pi_U"'] & & T^2Q \arrow[dl, "\pi_{T^2Q}"] \\ & TQ & \end{tikzcd} , \tag{16}$$

where

$$f = S \circ \pi_U + \mathrm{vl} \circ (\pi_U, F), \tag{17}$$

S is the geodesic spray of κ, and vl is the vertical lift (section 2). The following result explains the relationship of (16), (17) with the second-order control system (1), (2).

Lemma 1. For any $u \in \Gamma^0(\pi_U)$, a C^2 curve $t \mapsto q(t)$ in Q satisfies (14) if and only if it satisfies

$$\ddot{q} = f \circ u(\dot{q}). \tag{18}$$

Proof. This is immediate from the standard fact that, for any C^2 curve $t \mapsto q(t)$ in Q,

$$\ddot{q} = S(\dot{q}) + \mathrm{vl}(\dot{q}, \nabla_{\dot{q}}\dot{q}).$$

□

Lemma 2. For any $X \in \mathfrak{X}^0(Q)$ satisfying Assumption 2, there is a neighborhood $\tilde{W} \subset TQ$ of $X(Q)$ and $\tilde{Y} \in \mathfrak{X}(TQ)$ such that, for all $\varepsilon \in (0,1)$, $\varepsilon\tilde{Y}(\tilde{W}) \cap f(U_{\tilde{W}}) = \varnothing$.

Proof. Let W, $\mathcal{D}$, and Y be as in Assumption 2. We may assume that W is open. Let $\varphi\colon Q \to [0,\infty)$ be any smooth function such that $\varphi^{-1}(0) = Q \setminus W$. Define $\tilde{Y}_0, \tilde{Y} \in \mathfrak{X}^0(TQ)$ by

$$\tilde{Y}_0(v) = \varphi(\pi_{TQ}(v))\mathrm{vl}(v, Y(\pi_{TQ}(v)))$$

and $\tilde{Y} = S + \tilde{Y}_0$. Define $\tilde{W}_0 = \pi_{TQ}^{-1}(W)$, let $\tilde{W}_1 \subset TQ$ be any open neighborhood of $X(Q) \setminus \tilde{W}_0$ disjoint from 0_{TQ}, and define $\tilde{W} = \tilde{W}_0 \cup \tilde{W}_1$.

For any $\varepsilon \in (0,1)$, εS is not equal to S anywhere except on 0_{TQ}. Since εS and $S \circ \pi_U$ are respectively the horizontal parts of $\tilde{Y}$ and $S \circ \pi_U$, it follows that

$$\varepsilon\tilde{Y}(\tilde{W}) \cap f(U_{\tilde{W}}) \subset \tilde{W} \cap 0_{TQ} = \tilde{W}_0 \cap 0_{TQ}$$

for $\varepsilon \in (0,1)$. Thus, for $\varepsilon \in (0,1)$,

$$\varepsilon\tilde{Y}(\tilde{W}) \cap f(U_{\tilde{W}}) = \varepsilon\tilde{Y}_0(\tilde{W}_0 \cap 0_{TQ}) \cap f(U)$$

is empty if and only if $\varepsilon Y(W) \cap F(U_{T_WQ})$ is empty, which is the case since our assumptions imply that

$$\varepsilon Y(W) \cap F(U_{T_WQ}) \subset \varepsilon Y(W) \cap \mathcal{D}_W = \varnothing.$$

□

We now prove the second main result.

Proof of Theorem 2. Fix any $u \in \Gamma^0(\pi_U)$. Lemma 1 implies that (14) is uniquely integrable if and only if (18) is, and that $X(Q)$ is asymptotically stable for (14) if and only if it is the same for (18). Lemma 2 implies that the adversary condition (section 5.1) does not hold for (16), (17), so Theorem 3 implies that $X(Q)$ is not stabilizable for (16), (17). In particular, $X(Q)$ is not asymptotically stable for (18) and hence also not for (14). □

6. Discussion

In this paper, we established results determining whether prescribed vector fields on compact smooth manifolds, viewed geometrically as sections of the tangent bundle, can be asymptotically stabilized by general second-order control systems on the base manifold. We also illustrated the abstract results concretely in Examples 1, 2 and Figure 1.

For a broad class of fully actuated systems, we showed in Theorem 1 that the section corresponding to any smooth vector field can be made globally exponentially stable and normally attracting, which implies that the closed-loop second-order system asymptotically reproduces the prescribed first-order dynamics in a strong sense. Moreover, an explicit feedback law (9) to achieve this was provided by extending Koditschek's method [16, sec. 3.3] to our geometric setting.

In contrast, for a broad class of underactuated systems on manifolds of nonzero Euler characteristic, we showed in Theorem 2 that sections corresponding to "almost all" smooth vector fields cannot even be locally asymptotically stabilized. This includes, in particular, all vector fields having only isolated zeros. Here the underactuation requirement is that the control system takes values in a C^0 regular distribution of positive corank, a common situation for underactuated systems. Moreover, the image $F(U)$ actually need only be contained in a regular distribution over a neighborhood of the zeros of X, broadening the applicability of Theorem 2 (see Footnote 1).

It would be interesting to determine the optimal hypotheses under which the conclusions of Theorems 1 and 2 apply. For the latter, an extension of Coron's necessary condition for stabilizability of points in Euclidean spaces [41, Thm 2] introduced in [42, Thm 2] (see also [37, Thm 3]) might be useful, since it is stronger than the extension [22, Thm 3.2] of Brockett's condition [21, Thm 1.(iii)] used to prove Theorem 2 under some (but not all [9, Ex. 4]) conditions [9, Thm 4]. Additionally, it would be interesting to pursue geometric extensions of other techniques for embedding first-order dynamics in second-order systems distinct from Koditschek's method [43].

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